

Trainings ▾

Calibrate

Standard

⚠ WARNING ⚠

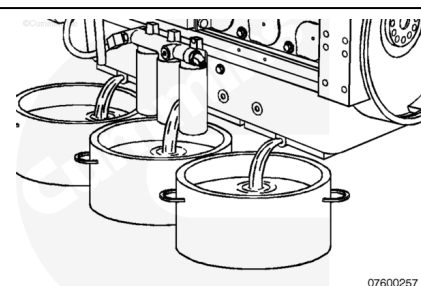
To reduce the possibility of personal injury, avoid contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.



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Note : For marine applications **only**, the lubricating oil dipstick can **not** be calibrated until the engine is in the vessel and the vessel is in the water.

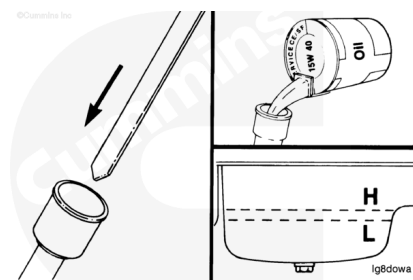
Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>) (Lubricating Oil System) in Section 8.

Allow five minutes for the oil to drain from the oil pan. If the engine and/or oil temperature is below 4°C [39°F], a longer period of time will be required for a full drain.

The engine or vehicle **must** be level to perform this procedure.

Install the dipstick in the dipstick tube housing.

Use the correct volume of clean 15W-40 oil. Fill the oil pan to the specified low or L oil level. Refer to Procedure 018-017 (</qs3/pubsys2/xml/en/procedures/20/20-018-017.html>) for the correct engine oil capacity for the application.

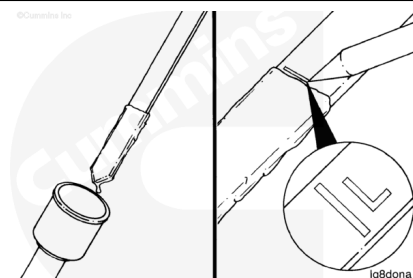


⚠ CAUTION ⚠

Exercise care when marking the dipstick, or the dipstick will break if the scribe mark is too deep.

Remove the dipstick, and scribe a mark across the stick at the oil level. Label the mark with an L to indicate the low oil level.

If a new, blank dipstick is being used, cut the dipstick approximately 38 mm [1.5 in] below the low or L oil level mark.

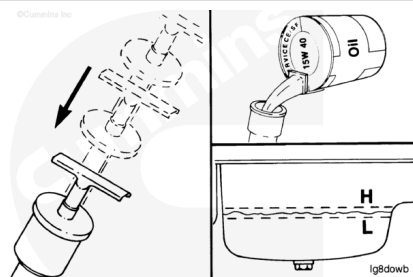


⚠ CAUTION ⚠

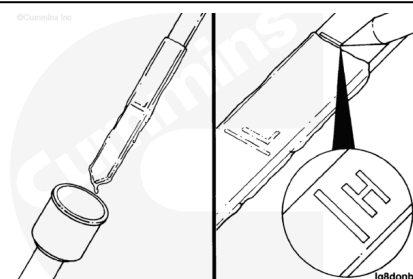
Only add the difference between low volume and high volume when adding oil to the specified high or low level. Excessive oil in the crankcase can cause premature engine failure.

Wipe the excess oil from the dipstick, and install it in the dipstick tube housing.

Use the correct volume of oil to fill the oil pan to the specified high or H oil level. Refer to Procedure 018-017 (</qs3/pubsys2/xml/en/procedures/20/20-018-017.html>) for engine oil capacity.



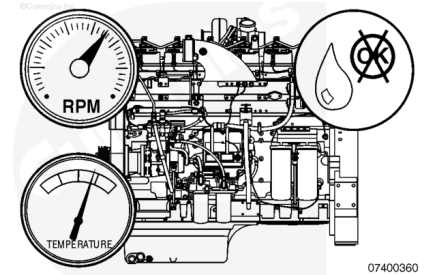
Remove the dipstick, and scribe a mark across the stick at the oil level. Label the mark with an H to indicate the high oil level.



Running Level Dipstick

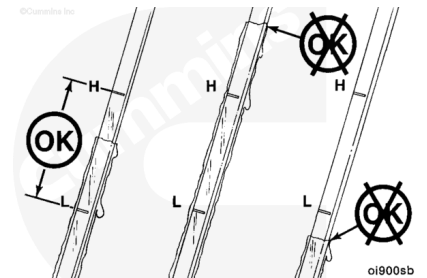
To calibrate the running level dipstick, the following procedure **must** be followed:

1. Check the engine oil level at static condition.
2. Operate the engine to bring the pan temperature up to approximately 99°C [210°F].
3. Add engine oil if necessary to bring engine oil level to the static HIGH level mark.
4. Return the engine to the low idle condition, disengage any engine driven accessories and put the transmission in neutral.
5. Allow to idle for 5 minutes.
6. Remove the oil filler cap.
7. Remove the dipstick from the tube and wipe clean.
8. Insert the dipstick fully and immediately withdraw.
9. Check the oil level marking on the engine idling side of the dipstick.
10. The running level readings **must** be between the high and low level range, on the engine idling side of the dipstick, and produce a clearly readable line. If not, repeat steps 4 and 5. The oil level indicated can be near to the HIGH level mark.



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9. Take further readings to confirm that readings are repeatable. However, it is **not** unusual for the level in the sump to vary slightly while the engine is idling. It can however remain near to the HIGH level mark. Total idling time can **not** exceed 10 minutes.
10. Install the oil filler cap.



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When satisfied that the dipstick registers the high running level correctly, it is suggested that the same test be run with the oil pan drained to the static low level mark.

At all times during the low level test, monitor the low oil pressure and temperature warning lamps and gauges.

Repeat the procedure used for the high level testing. The run level readings **must** be between the run high and low level range indicated on the dipstick. Total idling time **must not** exceed 10 minutes.

When satisfied that the dipstick markings are reflective of the behavior of the oil in the oil pan at low levels, add oil to the oil pan in equal steps until the high level is reached.

Each time an amount of oil is added check that the dipstick registers a change in the oil level. Each quantity of oil added can register a change in the level on the dipstick.

If it is practical and safe while the engine is idling, use the engine running dipstick procedure. If it is **not** possible, use the static level checking method.

As a guide, 6.4 mm [0.25 in] on the dipstick equates to approximately 3.8 liters [1 gal] of oil, if the high oil level is above the bottom of the oil pan adapter.

When the above tests have been completed and the operator is confident with the results, the vehicle or vessel can be returned to normal operations.

Last Modified: 30-Jun-2006
